Experiment No. 3

Title: Text Generation using a Pre-trained Transformer Model for a Question–Answer System.

Aim: To implement a text generation based Question–Answer system using a pre-trained Transformer model and generate answers to user questions.

Objectives:

1. To understand the basic concepts of text generation and Generative AI.
2. To understand how a pre-trained Transformer-based language model can be used for generating text.
3. To implement a simple Question–Answer system using Hugging Face Transformers and a pre-trained FLAN-T5 model.
4. To create an interactive Question–Answer interface using Gradio and analyze the generated responses.

Pre-Requisites:

1. Basic knowledge of Python programming.
2. Basic understanding of Deep Learning and neural networks.
3. Basic understanding of Natural Language Processing (NLP).
4. Familiarity with Python libraries and Google Colab.

Theory:

Text Generation: Text generation is a Generative AI task in which a model produces new text based on a given input or instruction. The generated text may be an answer, explanation, summary, dialogue, story, or other form of natural language output.

Transformer Models: Modern text-generation systems commonly use Transformer-based neural network architectures. Transformers process input text as tokens and learn relationships between tokens using attention mechanisms. Pre-trained Transformer models can be used for many Natural Language Processing tasks.

Pre-trained Language Models: Training a large language model from scratch requires very large datasets and substantial computational resources. In practical applications, a pre-trained model can be loaded and used directly for text generation. In this experiment, FLAN-T5 is used as a pre-trained instruction-following text-to-text model.

Question–Answer System: A Question–Answer system accepts a question as input and generates a natural-language answer. The workflow used here is: User Question → Prompt/Instruction → Tokenization → Pre-trained Transformer Model → Generated Tokens → Text Answer.

FLAN-T5: FLAN-T5 is an instruction-tuned version of the T5 family of text-to-text Transformer models. It can receive an instruction or question as text and generate a textual response. The experiment uses the google/flan-t5-base model.

Gradio Interface: Gradio is used to create a simple interactive user interface around the Python Question–Answer function. It allows the user to enter a question and view the generated answer.

Tools and Technologies Required

1. Google Colab
2. Python 3
3. PyTorch
4. Hugging Face Transformers
5. Hugging Face FLAN-T5 pre-trained model
6. Accelerate and SentencePiece
7. Gradio
8. GPU support (optional, but recommended for faster execution)

Procedure:

1. User enters a question.
2. An instruction/prompt is created for the model.
3. The tokenizer converts the input text into tokens.
4. The pre-trained FLAN-T5 model processes the tokens.
5. The model generates output tokens.
6. The tokenizer converts the generated tokens back into text.
7. The generated answer is displayed to the user through the Question–Answer interface.

Conclusion:

The experiment successfully demonstrated text generation using a pre-trained Transformer model to build a simple Question–Answer system. Students learned how a user question is converted into tokens, processed by a pre-trained generative model, and converted back into a natural-language answer. The experiment also demonstrated the use of Gradio for creating an interactive interface. This experiment provides a foundation for advanced Generative AI applications such as conversational chatbots, document Question–Answering and Retrieval-Augmented Generation (RAG) systems.